Manifold-regression to predict from MEG/EEG brain signals without source modeling

D. Sabbagh, P. Ablin, G. Varoquaux, A. Gramfort, D.Engemann

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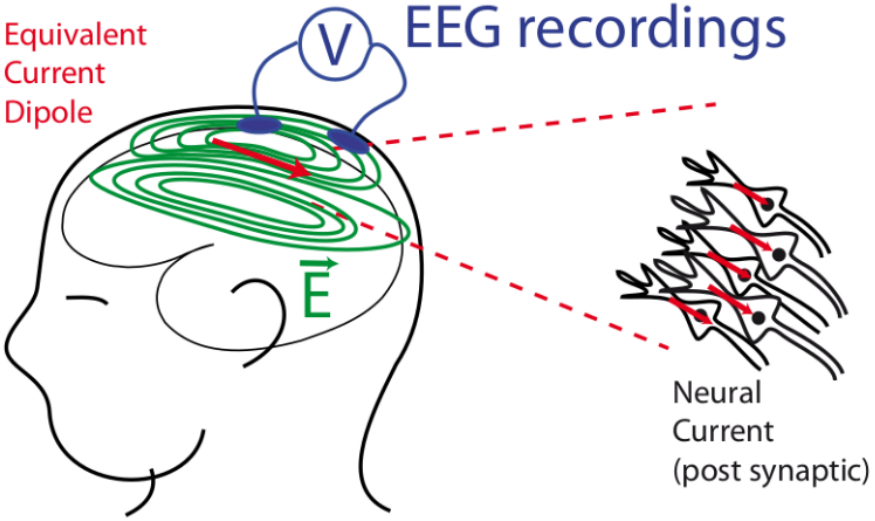
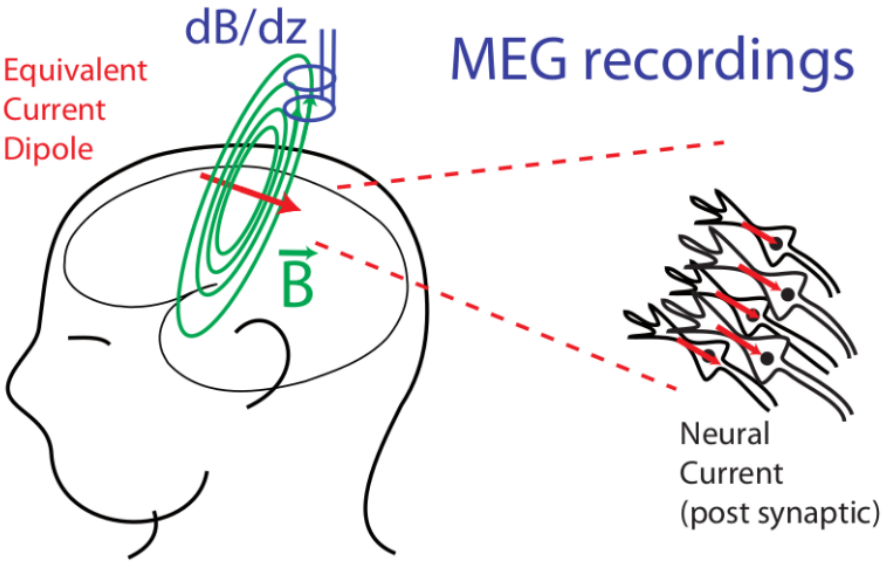
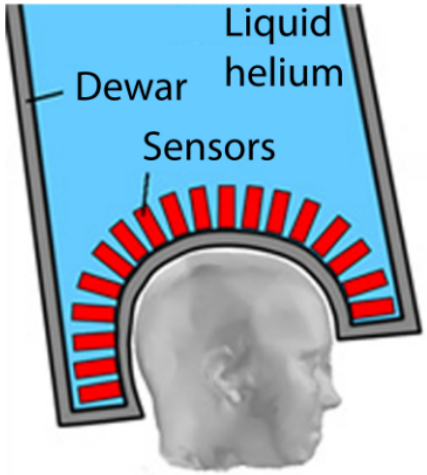




First EEG recordings in 1929 by H. Berger



Hôpital La Timone
Marseille, France

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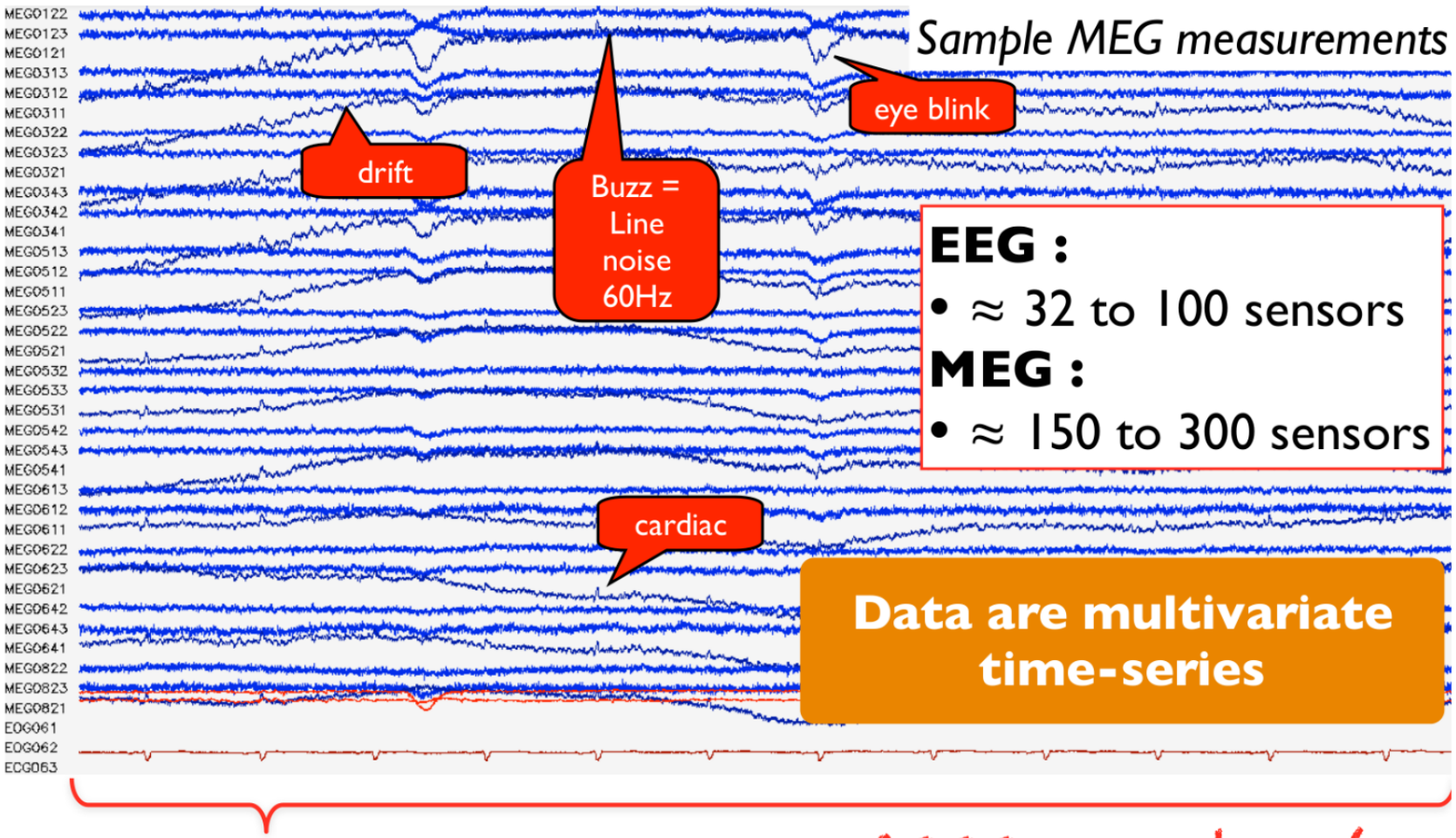

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We measure M/EEG signal of subject $i = 1 \dots N$ on P channels:

$$\boxed{\boldsymbol{x}_i(t) = \boldsymbol{A}\ \boldsymbol{s}_i(t) + \boldsymbol{n}_i(t)} \in \mathbb{R}^P \quad t = 1 \dots T$$

- *mixing matrix* $\boldsymbol{A} = [\boldsymbol{a}_1, \dots, \boldsymbol{a}_Q] \in \mathbb{R}^{P \times Q}$ fixed accross subjects
- *source patterns* $\boldsymbol{a}_j \in \mathbb{R}^P, j = 1 \dots Q$ with $Q < P$
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Under stationnarity and gaussianity assumptions, we can represent band-pass filtered signal by its second-order statistics

$$\boldsymbol{C}_i = \mathbb{E}[\boldsymbol{x}_i(t)\boldsymbol{x}_i(t)^\top] \simeq \frac{\boldsymbol{X}_i \boldsymbol{X}_i^\top}{T} \in \mathbb{R}^{P \times P} \quad \text{with} \quad \boldsymbol{X}_i \in \mathbb{R}^{P \times T}$$

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We want to predict a continuous variable: $\boxed{y_i = \sum_{j=1}^{Q} \alpha_j f(p_{i,j})} \in \mathbb{R}$

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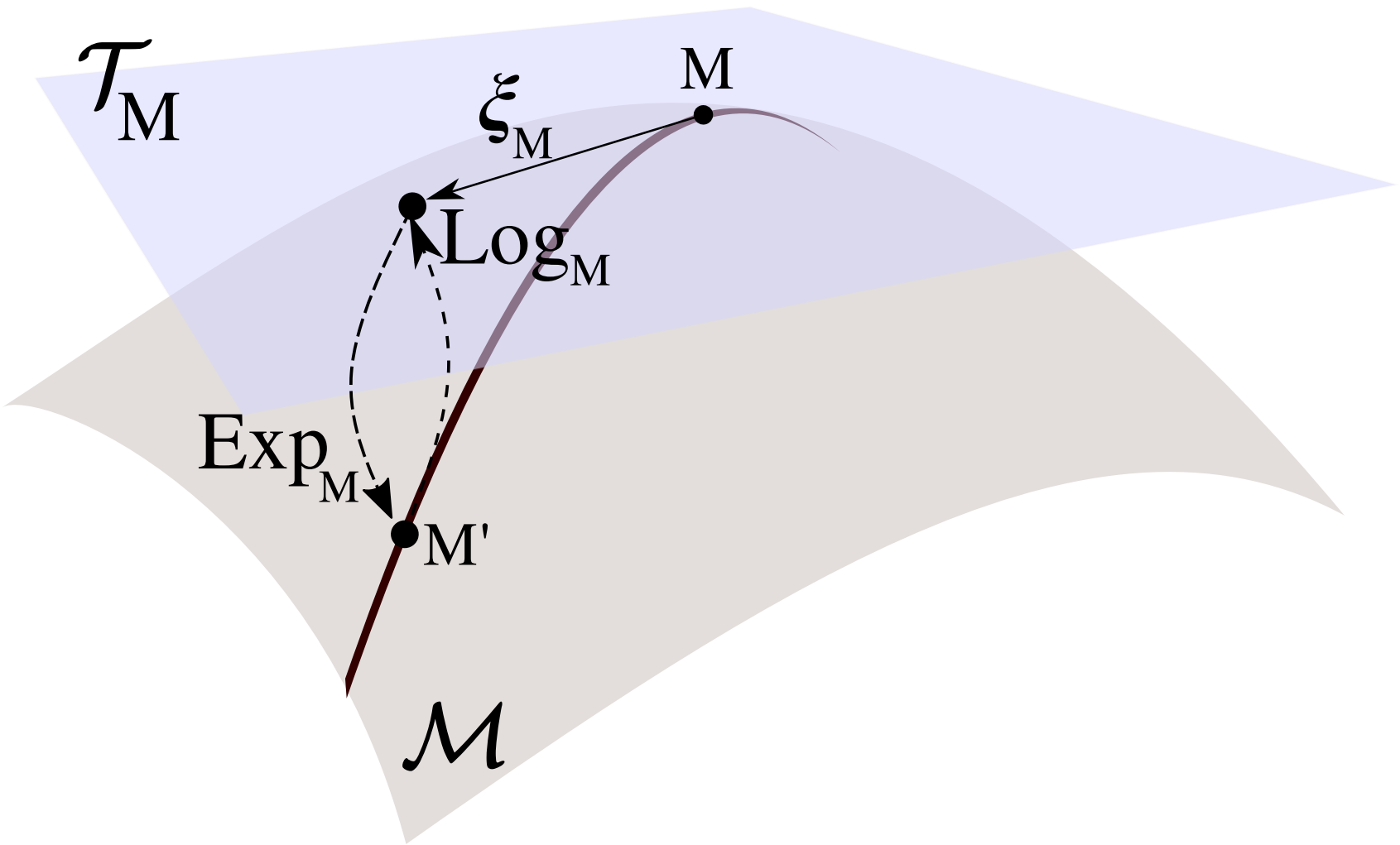
  $\boldsymbol{C}_i$ live on a Riemannian manifold so can't be naively vectorized

Riemannian matrix manifolds (in a nutshell)



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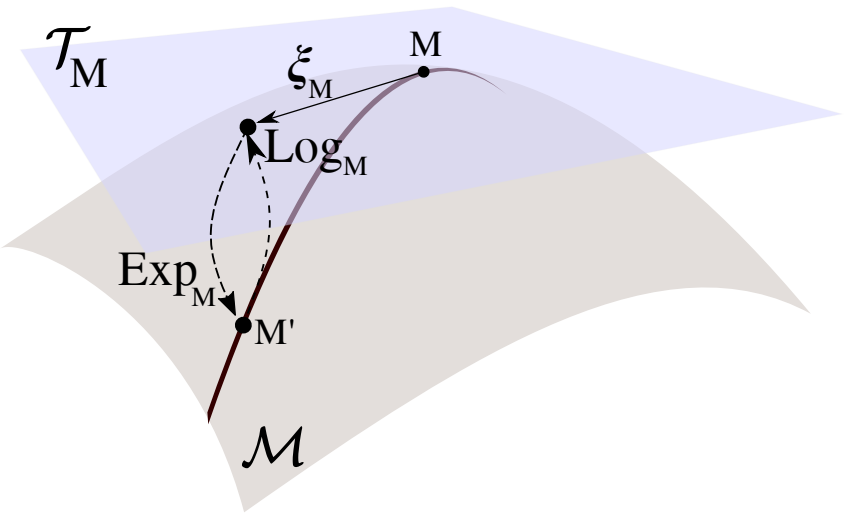

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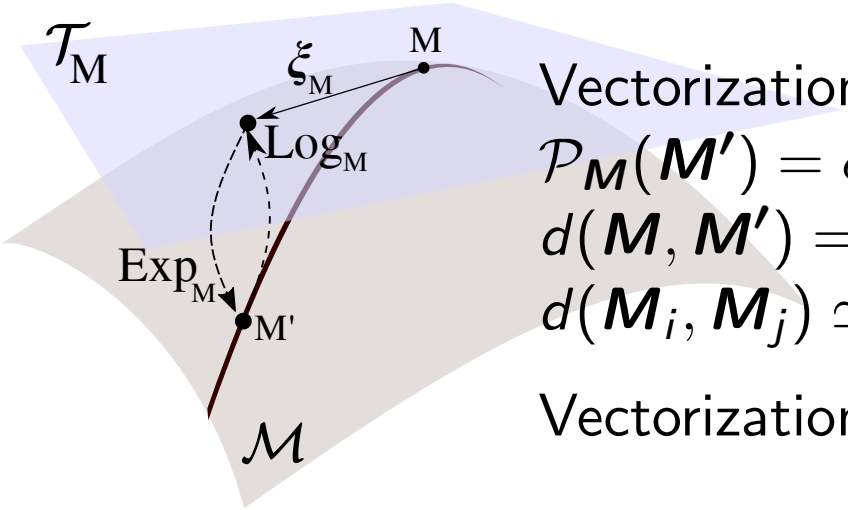

Vectorization operator:

$$\mathcal{P}_{\boldsymbol{M}}(\boldsymbol{M'}) = \phi_{\boldsymbol{M}}(\mathrm{Log}_{\boldsymbol{M}}(\boldsymbol{M'})) \simeq \mathrm{Upper}(\mathrm{Log}_{\boldsymbol{M}}(\boldsymbol{M'}))$$

$$d(\boldsymbol{M}, \boldsymbol{M'}) = \|\mathcal{P}_{\boldsymbol{M}}(\boldsymbol{M'})\|_2 + o(\|\mathcal{P}_{\boldsymbol{M}}(\boldsymbol{M'})\|_2)$$

$$d(\boldsymbol{M}_i, \boldsymbol{M}_j) \simeq \|\mathcal{P}_{\overline{\boldsymbol{M}}}(\boldsymbol{M}_i) - \mathcal{P}_{\overline{\boldsymbol{M}}}(\boldsymbol{M}_j)\|_2$$

Vectorization operator key for ML

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- use those vectors as features in regularized linear regression algorithm (e.g. ridge regression) with parameters $\boldsymbol{\beta} \in \mathbb{R}^K$ assuming that $y_i \simeq \boldsymbol{v}_i^\top \boldsymbol{\beta}$

Distance and invariance on positive matrix manifolds

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Affine invariance is key: working with $\boldsymbol{C}_i$ is then equivalent to working with covariance matrices of sources $\boldsymbol{s}_i$

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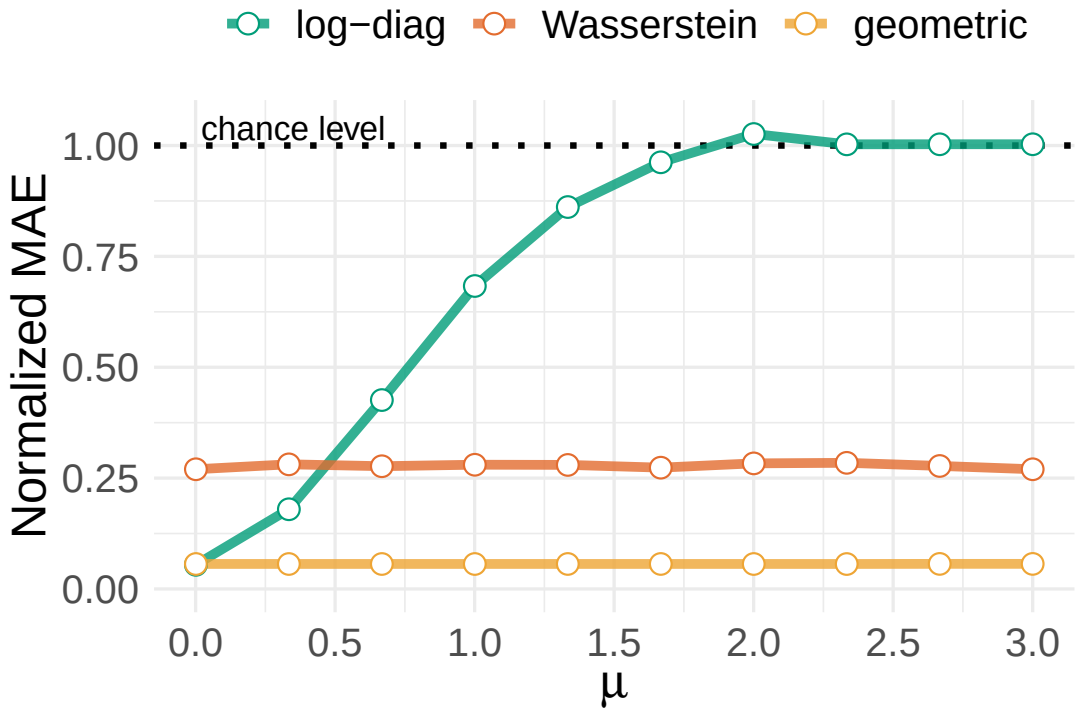
We generate i.i.d. samples following the log linear generative model. $\boldsymbol{A} = \exp(\mu \boldsymbol{B})$ with $\boldsymbol{B} \in \mathbb{R}^{P \times P}$ random



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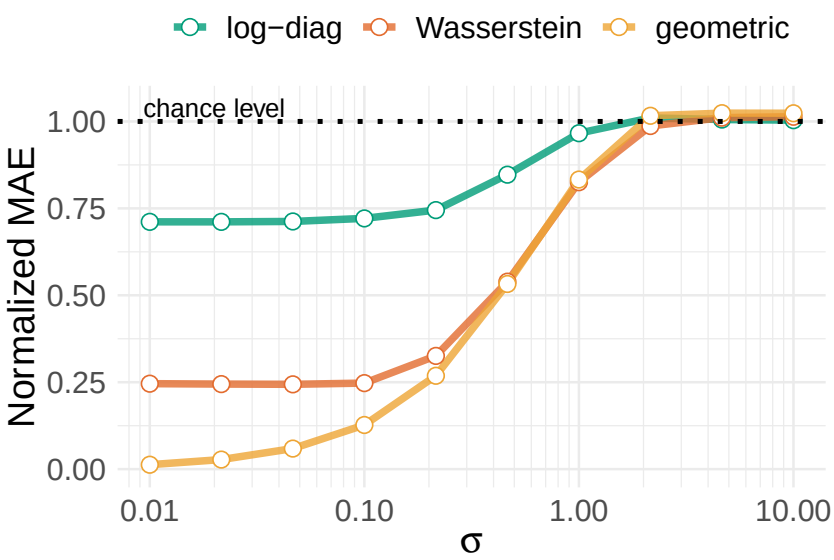

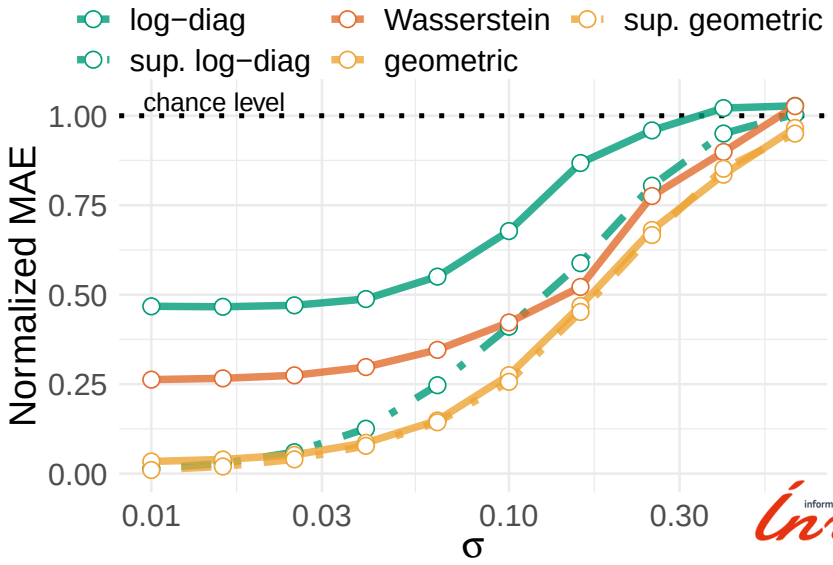

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- Rank-deficient signals (*e.g.* cleaning process): $\boldsymbol{C}_i \in \mathcal{S}^+_{P,R}$

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`https://arxiv.org/abs/1906.02687`